

Bibek Kumar Tamang

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Summary

Software Engineer with experience in Java and Spring Boot creating scalable applications with strong foundation in data structures, algorithms, and system design. Proficient in AWS cloud technologies with proven ability to architect and implement high-performance systems that scale reliably while following modern development best practices.

Skills

- **Programming Language:** Java | SQL | JavaScript | C++ | Python
- **Cloud & DevOps:** AWS | Docker | Git | Kubernetes | CI/CD
- **Backend & Frameworks:** Spring Boot | REST APIs | JWT Authentication | Microservices | React
- **Databases:** MySQL | PostgreSQL | MongoDB | Redis
- **Core Competencies:** Data Structure & Algorithms | System Design | Full-Stack Development
- **Operating System:** Windows, Linux, MacOS

Experience

Dallo Tech

Jan 2025 - Jun 2025

Software Developer, Intern

Lalitpur, Nepal

- Built and extended backend REST APIs for a multi-vendor e-commerce platform using **Java and Spring Boot**, supporting core workflows such as product catalog management, order processing, and vendor-specific access control.
- Implemented **JWT-based authentication and role-based authorization**, defining access boundaries for admin, seller, and customer roles, and ensuring secure request handling across protected endpoints.
- Designed and maintained **relational database schemas** using PostgreSQL/MySQL, including table normalization, indexing of frequently queried fields, and writing optimized SQL queries to improve read performance and maintain data consistency.
- Worked on backend deployment by **containerizing services with Docker** and deploying them to **AWS EC2**, while integrating **AWS S3** for static asset storage and gaining exposure to cloud-based CI/CD workflows.
- Collaborated with senior developers during code reviews and sprint planning, contributing bug fixes, feature enhancements, and refactoring efforts while following Agile development practices and Git-based version control.

Selective Projects

High-Throughput Ticket Reservation System | [High-Throughput Ticket Reservation System](#)

- Built a backend ticket reservation service using Spring Boot and PostgreSQL to manage seat availability.
- Used Redis for basic locking to prevent double booking during concurrent purchase attempts.
- Implemented rate limiting and error handling to protect APIs from misuse.
- Deployed the service using Docker and tested concurrent access using load testing scripts

Real-Time Chat Application with Spring Boot & React | [Real-Time Chat Application with Spring Boot & React](#)

- Developed a real-time messaging backend using Spring Boot WebSocket and STOMP
- Stored chat messages in MongoDB and managed active user sessions
- Built a React frontend to send and receive messages in real time
- Containerized backend, frontend, and database using Docker Compose

Maze Solver | [Maze Solver](#)

- Developed a full-stack maze generation engine using Spring Boot microservices and REST APIs, implementing modular pathfinding algorithms (BFS/DFS) with 99.9% accuracy.
- Designed maze-solving algorithms with optimized data structures (priority queues, adjacency lists), achieving $O(n)$ average-case complexity on 64×64 grids and reducing rendering latency by 30% via efficient path-drawing.
- Implemented a solo DevOps workflow using Git, Maven, and Docker, achieving 100% build reproducibility and enabling seamless deployment from local development to cloud hosting environments.

Education

Youngstown State University

May 2026

Bachelor of Science, Computer Science

- **Coursework:** Data Structure & Algorithm, Operating System, Object Oriented Programming, Data Science & Machine Learning, Computer Architecture, Networking Concepts & Administration, Development of Database,
- Calculus I, Calculus II, Calculus III, Probability & Statistics, Linear Algebra & Matrix Theory